Ali Wael Mohamed Ali Ahmed

Address: Dokki, Giza, Egypt | **Phone:** 01011050393 | **Email:** aliwael908@gmail.com | **LinkedIn:** linkedin.com/in/ali-wael908

Professional Summary

Passionate and curious Mechatronics Engineering Senior driven to take on new challenges, continuously learn, and contribute to purposeful, interesting projects. Strong foundation in **Robotics (ROS2, AI for Robotics)** , **Embedded C/C++ development** , **low-level driver implementation** (HAL, CAN, I2C, ADC, PWM), **Machine Learning**, and **Control Theory** . Experienced in bridging physical constraints with efficient control logic across Robotics, Medical Devices, and electromechanical systems. Highly adept at leveraging Generative and Agentic AI tools to optimize workflows.

Education

Cairo University, Giza, Egypt B.Sc. in Mechatronics Engineering — Graduated: July 2026

Relevant Coursework: Control Systems, Automatic Control, Microprocessors & Microcontrollers, Thermodynamics, Embedded Systems, Advanced Robotics, Design of Mechatronics Systems, Electric Drive, Pattern Recognition & Artificial Neural Networks, Machine Intelligence.

Professional Experience

Embedded Linux Engineer Intern | Ezzmedical, Cairo, Egypt (Aug 2025 – Sep 2025)

Developed and deployed a highly reliable, custom Linux operating system for EZvent, the first Egyptian-developed medical ventilator to achieve unconditional regulatory approval.

Ensured system-level stability for life-critical operations, aligning with strict medical device compliance standards. Validated hardware-software integration by deploying a Qt-based interface onto ventilator boards, optimizing boot processes and low-level system performance.

Level 2 Automation Engineer Intern | Al Ezz Dekheila Steel Co. (EZDK), Alexandria, Egypt (Jul 2024)

Mapped complex industrial data flows between Level 1 and Level 2 automation hierarchies. Documented real-time data acquisition strategies to improve communication clarity across distributed industrial systems.

Technical Projects

Machine Learning Enhanced Emergency BVM Ventilator (Team Lead) (Sep 2025 – Present)

Embedded C/C++ & Architecture: Designed Software Architecture Document & Software Requirements Specifications documents.Developed robust, safety-critical ventilation firmware in C/C++, managing real-time sensor calibration, 3 layer Architecture, serial communication, sensor filters and critical safety alarms. **Electrical Analysis, Integration & Wiring:** Designed the full electrical system making sure of proper integration and compatibility between all components. Assembled all electrical components and accomplished a stable and safe wiring for the full electrical system. **Electromechanics & Motor Control:** Implemented advanced S-Curve motion profiling for stepper motors to ensure jerk-free operation, translating physical constraints and torque limits into efficient code.

Low-Level Drivers & Signals: Designed sophisticated signal conditioning for pressure (MPX5010DP) and environmental (BMP280) sensors using ADC oversampling and Exponential Moving Average (EMA) filters to mitigate noise and drift. **Cross-Functional Agile Leadership:** Directed an interdisciplinary team (mechanical, electrical, software, medical), managing constraints like cooling strategies and electrical integration, while acting as Scrum Master.

Automated Bottling & Capping Line Simulation (Dec 2024 – Jan 2025)

PLC Programming & Automation: Programmed Siemens PLCs in TIA Portal to simulate full bottling and capping operations with safety interlocks and reset logic.

HMI & Process Monitoring: Designed a comprehensive HMI interface for real-time process monitoring and simulation of manual interventions, fully documenting industrial logic.

Robotic Arm Trajectory Automation & Teleoperation (Nov 2025 – Dec 2025)

CAN Bus Integration: Configured low-level **CAN bus** network interfaces in Linux environments to monitor live telemetry and handle real-time connection states. **Control Laws:** Translated physical mechanics into software control laws, utilizing the robot's "teach mode" for high-precision trajectory tracking and HTTP requests for smartphone IMU teleoperation.

Minimalist Smartwatch on STM32 Microcontroller (Jun 2024)

Microcontroller Architecture: Built a real-time digital watch programmed entirely in **bare-metal ARM Assembly** on an STM32F103C8T6. **Deep Hardware Understanding:** Demonstrated deep understanding of microcontroller architecture including memory mapping, direct register manipulation, interrupts, and hardware timers to achieve high accuracy. Interfaced a TFT display with optimized GPIO protocols.

Pick and Place Robotic Arm & Custom Gripper Control (April 2026 – May 2026)

Closed-Loop PID Control: Engineered a custom robotic gripper mechanism driven by a lightweight (10g) open-loop DC motor. Implemented a closed-loop PID control system utilizing potentiometer feedback for precise position regulation and motor control. **Motion Profiling & Kinematics:** Developed synchronized multi-axis interpolation algorithms for the robot's servo motors, ensuring fast, smooth, and vibrationfree movement profiles.

Machine Fault Recognition using CNN (Mar 2026 – May 2026)

Deep Learning for Diagnostics: Developed an end-to-end Convolutional Neural Network (CNN) model for machine fault recognition. **ML Pipeline Implementation:** Executed comprehensive data preprocessing, model training, and inference testing, demonstrating capabilities directly applicable to predictive maintenance and anomaly detection.

Autonomous Museum Guide Robot & AI Integration (Feb 2025 – May 2025)

Mechanical Design: Designed the car chassis in SOLIDWORKS specifically for laser cutting. The design incorporated several flat parts to ensure we could utilize a single sheet of acrylic for cost efficiency and manufacturing speed. I designed the parts to have finger joints compatible with each other for easy assembly, and ensured proper holes were mapped out for the electronic components and the PCB.

Control & Sensor Fusion: Integrated an MPU6050 IMU and encoders, applying PID control loops for stable, real-time autonomous navigation & Go To Goal logic, applying Aruco Markers Detection and Obstacle Avoidance using phone's camera and real time obstacle detection.

Machine Learning & Final Delivery:

- Deployed YOLOv11 object detection, enhancing real-time recognition latency by 30%.
- Trained a custom YOLOv11 model on 3 specific museum artifacts using 50+ images per artifact covering different angles and lighting.
- When the robot detects an artifact, it stops and utilizes a Text-to-Speech (TTS) integration to present the artifact to tourists.

Core Skills

- **Programming & Scripting:** C/C++, Python, ARM Assembly, MATLAB / Simulink.
- **Embedded Systems:** Embedded Linux, U-Boot, Kernel menuconfig, Microcontroller Architecture (STM32, ESP32, Arduino), Low-Level Software, Hardware Abstraction Layers (HAL), ADC, PWM, SPI, I2C, CAN Bus.
- **Control & Automation:** Control Theory (Automatic Control, PID, Kinematics), Motor Control & Motion Profiling, Electronics, Mechanics & Thermodynamics, PLC Programming (TIA Portal).
- **AI & Machine Learning:** Actively Exploring Agentic AI frameworks to automate C/C++ development and debug complex hardware/software issues. Practical experience building end-to-end ML pipelines (CNNs, Bayes Classifiers, OpenCV, YOLO) for fault recognition and predictive analytics.
- **Standards & Quality:** Familiarity with high code quality standards, strict compliance mindsets (Medical/Automotive), Agile methodologies.
- **Tools:** Keil, TIA Portal, Git, GitHub, SolidWorks, ROS, Linux, Matlab.

Certifications & Languages

Certifications: Embedded Linux Course (IMT, 2023) | DELF B2 (French Ministry of Education, 2021) **Languages:** Arabic (Native), English (Proficient), French (Advanced/B2), Spanish (Beginner)